# Aylin Davoudi

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## PROFILE

Aerospace engineering master's student with a multidisciplinary background across diverse engineering projects, specializing in propulsion and fluid mechanics. Seeking a 1–2 month summer internship.

## EDUCATION

### Master of Aerospace Engineering – ISAE-SUPAERO, Toulouse, France
### Advanced Aerodynamics & Propulsion Major

**Relevant Coursework:** *Rocket Propulsion, Space Systems, Orbital Mechanics, Acoustics, Computational & Experimental Fluid Dynamics, Applied Aerodynamics, Embedded Systems, Information Engineering, Electromagnetism applied to Avionics, Flight & Ground Loads, Optimization, Flight Dynamics, Algorithms & Computing, Scientific Computing and Programming, Control Engineering, Airworthiness, Systems Engineering* *2025–2027*

### International Bachelor in Mechanical, Materials & Aerospace Engineering – INSA Lyon

**Relevant Coursework:** *Fluid Mechanics, Heat Treatments, Material Science & Structural Analysis, Composite Materials, Machine Elements, Transducers & Measurements, Flight Dynamics & Control, Mechanical Design, Manufacturing* *2022–2025*

**Courses at ECAM LaSalle (partnership with IBMMAE, INSA Lyon):**
*Compressible Flows & Propulsion, Fluid Mechanics, Thermodynamics, Manufacturing & CNC Machining)* *2022–2025*

**Exchange Semester – University of Twente, The Netherlands:***Fluid Mechanics 2, Heat Transfer, Control and Mechatronics* *FEB– JUN 2024*

**High School Diploma – Georgian American International High School, Tbilisi, Georgia**
GPA: 4.0 *2020–2021*

## PROJECTS

**Droplet Evaporation in Real-Gas & Supercritical Conditions – Master's Research Project, ISAE-SUPAERO** *2025–2027*

- Extended a Python-based evaporation solver ($D^2$-law, Abramzon–Sirignano) to real-gas conditions by implementing a Peng–Robinson equation of state with multi-component mixing rules.
- Applied fugacity-based corrections to thermodynamic properties (vapour pressure, latent heat, heat capacity) to model high-pressure environments.
- Investigated transcritical and supercritical regimes where classical evaporation models fail, with applications to liquid rocket engine injectors. *(Ongoing)*

**Design & Optimization of a Pintle Injector – Bachelor Thesis, INSA Lyon** *2024–2025*

- Designed and modeled a RP-1/LOX based pintle injector for lander propulsion, developing analytical models for mass flow, momentum ratio, and combustion performance.
- Performed CFD simulations (Star-CCM+) to evaluate the influence of pintle angle and gap on mixing efficiency and thrust.
- Built and tested a physical prototype, validating results against analytical predictions and CFD simulations.

**Design & Optimization of Regenerative Cooling Channels for Vulcain 2 – University of Twente** *2024*

- Designed and optimized regenerative cooling channels for a rocket nozzle using analytical methods and CFD to maintain safe wall temperatures.
- Analyzed flow and heat transfer to refine channel geometry and improve cooling efficiency.
- Validated performance through wind tunnel experiments, comparing heat transfer measurements with theoretical predictions.

**Mechatronic Design of a Mirror Manipulator for Inter-Satellite FSO Communication – University of Twente** *2024*

- Designed a precision mirror positioning system, defining tracking requirements and implementing a PID controller with feedforward control.
- Modeled and simulated system dynamics (Spacar, Simulink); designed compliant flexures in SolidWorks with safety validation; built and tested the prototype.

- Achieved lowest tracking error among all groups ($< 30\,\mu$rad).

**Ultrasound-Based NDT for Aircraft Maintenance – INSA Lyon** *2024*

- Designed and simulated ultrasonic sensors in COMSOL for defect detection in aerospace structures, optimizing piezoelectric configurations and placement.
- Analyzed wave propagation in composite materials to improve detection of cracks, delamination, and voids for predictive maintenance.

**Aerodynamic Design & Assessment of the FF72 Aerial Firefighting Aircraft – ISAE-SUPAERO** *2025–2026*

- Designed wing configuration through airfoil screening and aerodynamic analysis (XFLR5) to improve efficiency and reduce induced drag.
- Modeled the FF72-Scooper (ATR72 with floaters) in OpenVSP; quantified parasitic drag increase and its impact on several stages of flight.
- Assessed stability, ground effect, and CS-25 constraints; proposed design improvements achieving a 14% reduction in total drag.

**Information Engineering for a Mars Imaging Mission – ISAE-SUPAERO** *2025–2026*

- Applied information theory to estimate compression limits and channel capacity for a Mars rover communication link; implemented a JPEG-based compression algorithm in Python (Huffman coding, quantization tuning) and evaluated performance trade-offs.
- Exploited a one-time pad vulnerability in an LFSR-based stream cipher by XOR-ing three images encrypted with the same seed to recover visual information about Martian weather conditions.

**Fabrication, Characterization & Simulation of Unidirectional Composites – INSA Lyon** *2024*

- Fabricated composite samples using contact molding and vacuum techniques; performed mechanical characterization (tensile tests, DIC, vibratory analysis).
- Simulated mechanical behavior in Abaqus to analyze stress response and optimize stacking sequences.

**High Performance Computing & Finite Element Methods – ISAE-SUPAERO** *2025–2026*

- Implemented MPI parallelization (Fortran/C) for a Jacobi solver and solved Laplacian problems using GetFEM++ with Gmsh meshes (Python); analyzed scalability and finite element convergence.

## WORK EXPERIENCE

**Thermal Cooling Team Member — PERSEUS (CNES) / ISAE SUPAERO** *2026 – present*

- Designing regenerative cooling circuits for 15 kN LOX–ethanol rocket engines.

**Engineering Intern — CMD Gears, Cambrai, France** *Apr 2025 – Aug 2025*

- Modeled conical interference fits to evaluate contact pressure, material behavior, and torque transmission (FEMAP, Solid Edge, KISSsoft/KISSsys).
- Investigated lubrication effects and conducted analytical/FEM parametric studies; contributed to bearing design, gear meshing, and structural analyses.
- Delivered technical reports and engineering justifications for internal and client projects.

**Assembly Line Worker — Stellantis, Poissy, France** *Jun – Aug 2023*

- Performed assembly operations (kitting, leak testing, wheel and bumper assembly) on automotive production lines.
- Developed precision, adaptability, and teamwork in a fast-paced manufacturing environment.

## SKILLS

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| **Programming** | Python, C/C++, Fortran |
| **Software** | MATLAB/Simulink, SolidWorks, Solid Edge, Fusion 360, FEMAP, KISSsoft/KISSsys, Ansys, Abaqus, Star-CCM+, COMSOL Multiphysics, XFOIL/XFLR5, OpenVSP, IBM Engineering Rhapsody (SysML), SU2, GetFEM++, Gmsh, ParaView, Microsoft Office, ISAE softwares: Longisim, Transsim. Timeus |
| **Machining** | Certified HAAS lathe and mill operator, G-code programming |
| **Languages** | English (IELTS Academic 8.5), French (intermediate), Spanish (intermediate), Portuguese (beginner), Russian (beginner) |